

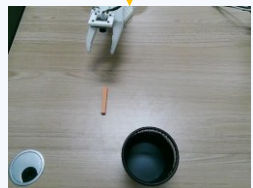
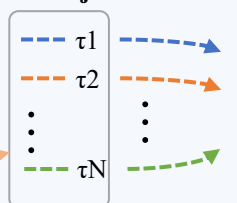
(a) Sparse Demonstrations

Multimodal Observations

Teleoperation



Trajectories



Front View



Wrist View

Vision-Language-Action Model

Train VLA Model



Limited Data, Low Accuracy



Limited Data, insufficient Coverage & Suboptimal Policy

(b) Structure-aware 3D Gaussian Scene Representation

Object-level Modeling

Multi-View Real Scene



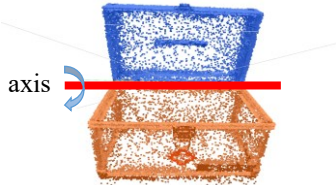
Decouple the Object Models



Decouple the Background Models



Add Hinge Constraint



Combine In Simulation



- ✓ Object-Decoupled
- ✓ Background-Decoupled
- ✓ Structural Constraints
- ✓ Physics-aware

(c) Policy Enhancement Loop

Object-level Modeling

Sparse Demo Inference in Simulation



Collect valid data



Dataset Augmentation

Reality Dataset
N samples

+

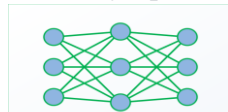
Virtual Dataset
M samples

→

Enhanced Dataset
M+N samples

Policy Update

Training



Inference

Data collection